Statistics of Financial Markets

Seminar 1: Returns and Stationarity

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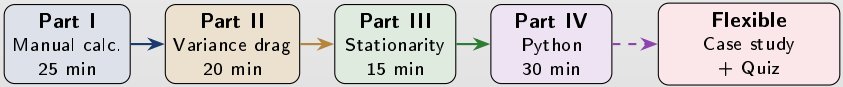
Learning Objectives — Seminar 1

By the end of this seminar, you will be able to:

1. **Compute** simple and logarithmic returns from price data
2. **Compare** the temporal aggregation of both return types
3. **Explain** variance drag and its impact
4. **Argue** why we analyse returns rather than prices (stationarity)
5. **Implement** return calculations in Python



Total: 90 min lecture + case study and quiz as supplementary material

Exercise 1: Computing simple and log returns

Price data

t	0	1	2	3
P_t	100	110	99	120

$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$ (simple)

- $R_1 = 10/100 = +10.00\%$
- $R_2 = -11/110 = -10.00\%$
- $R_3 = 21/99 = +21.21\%$

$r_t = \ln(P_t/P_{t-1})$ (logarithmic)

- $r_1 = \ln(1.10) = +9.531\%$
- $r_2 = \ln(0.9\overline{0}) = -10.536\%$
- $r_3 = \ln(1.2121) = +19.237\%$

Key observation

$R_1 + R_2 = 0\%$, but the price dropped to 99! Simple returns are **not additive** — they compound multiplicatively: $\prod(1 + R_t)$.

Exercise 1: Additivity and comparison

Verification: log additivity

- $r_1 + r_2 + r_3 = +18.232\%$
- $\ln(P_3/P_0) = \ln(1.20) = +18.232\%$ ✓

	Per. 1	Per. 2	Per. 3	**Total**
R_t	+10.00	−10.00	+21.21	+20.00
r_t	+9.53	−10.54	+19.24	+18.23

Units: %. Both methods ⇒ same terminal wealth.

Simple compounding (multiplicative)

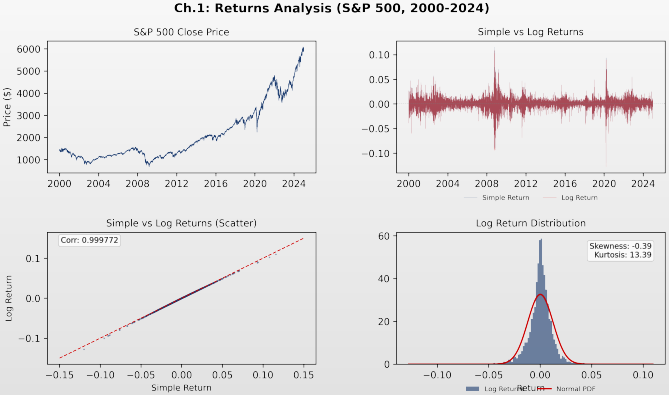
$$\prod_{t=1}^{3}(1 + R_t) = 1.10 \times 0.90 \times 1.2121 = 1.20$$

- $R_{\text{total}} = 20\%; \quad e^{r_{\text{total}}} - 1 = 20\%$ ✓

Conclusion

- Log returns **add up**, simple returns **multiply**
- We use log returns in statistical modelling for temporal additivity

Visualisation: Return analysis (S&P 500, 2000–2024)



- **Scatter**: correlation ≈ 0.9998 between R_t and r_t — nearly identical for small values
- **Histogram**: heavier tails than Normal ($K = 13.4$, $S = -0.39$)

SFM_ch1_returns

Quick poll — Vote now!

Question

- You invest **$100** in an asset. It rises by $+50\%$ in one year, then falls by -50% the next year.
- **How much are you left with?**

A) $100 (0% net)

B) $90 ($-10\%$)

C) $75 ($-25\%$)

D) $50 ($-50\%$)

Quick poll — Answer

Calculation

- $\$100 \times 1.50 = \$150$, then $\$150 \times 0.50 =$ **$75**
- You lose **25%**, not 0%!

Why?

- The 50% gain applies to $100, but the 50% loss applies to $150 — a *larger base*
- This is **variance drag** in action: volatility *destroys* wealth
- Formula: loss $\approx \sigma^2/2$ per year (we prove this in Part II)

Guided discussion: Why don't +50% and −50% cancel out?

You just saw: \$100 → \$150 → \$75. But what about log returns?

Simple returns (multiplicative)

$$(1.50)(0.50) = 0.75 \neq 1.00$$

- $R_1 + R_2 = +50\% + (-50\%) = 0\%$
- But terminal wealth: \$75, not \$100!
- Simple returns are **not** additive

Log returns (additive)

$$r_{\text{up}} = \ln(1.50) = +40.55\%$$

$$r_{\text{down}} = \ln(0.50) = -69.31\%$$

- Sum: $-28.77\% = \ln(0.75)$ ✓
- Log returns **correctly** reflect the loss

Key message

- The asymmetry is dramatic: you need +100% to recover from −50%
- The higher the volatility, the more severe the loss

Variance Drag: Formula and intuition

Definition 1 (Variance drag)

- Relationship between the expected arithmetic and geometric return:

$$\mathbb{E}[r] \approx \mathbb{E}[R] - \frac{1}{2}\sigma^2$$

- $\mathbb{E}[R]$ — expected simple return
- $\mathbb{E}[r]$ — expected log return
- σ^2 — variance of simple returns

- Variance drag grows **quadratically** with volatility: doubling $\sigma \Rightarrow$ quadrupling the loss
- This is a consequence of **Jensen's inequality**: $\mathbb{E}[\ln(1+R)] < \ln(1+\mathbb{E}[R])$ since ln is concave
- **Implication**: volatility *destroys* wealth in the long run, even if the arithmetic return is positive

Exercise 2: Two assets, same mean, different volatility

Problem

Asset	$\mathbb{E}[R]$	σ	$\mathbb{E}[r] \approx \mathbb{E}[R] - \sigma^2/2$
A	12%	10%	?
B	12%	30%	?

Solution

- **Asset A**: $\mathbb{E}[r]_A \approx 0.12 - 0.5 \times (0.10)^2 = 0.12 - 0.005 = 11.50\%$
- **Asset B**: $\mathbb{E}[r]_B \approx 0.12 - 0.5 \times (0.30)^2 = 0.12 - 0.045 = 7.50\%$

Question for students

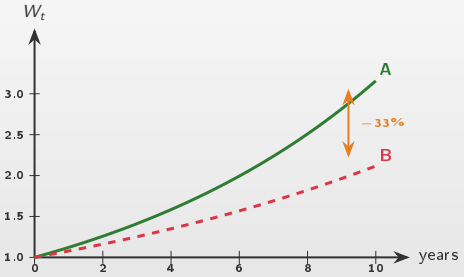
- Which asset produces more wealth in the long run?
- Compute W_{10} for both ($W_0 = 1$)

Exercise 2: Solution — Long-run wealth

$$W_T = W_0 \cdot e^{T \times \mathbb{E}[r]}$$

- **A**: $W_{10} = e^{10 \times 0.115} = 3.16$ (tripled)
- **B**: $W_{10} = e^{10 \times 0.075} = 2.12$ (doubled)

Asset	$\mathbb{E}[R]$	σ	$\mathbb{E}[r]$	W_{10}
A	12%	10%	11.50%	3.16
B	12%	30%	7.50%	2.12



Key conclusion

- **High volatility destroys wealth**! Asset B has a drag of 4.5%/yr — over 10 years, it costs ≈33% of wealth
- *"Volatility is the cost of uncertainty."*

Exercises 3–5

Exercise 3: Additivity of log returns — Show that $r_{0,T} = \ln(P_T/P_0) = \sum_{t=1}^{T} r_t$

Hint: write $\frac{P_T}{P_0} = \frac{P_1}{P_0} \cdot \frac{P_2}{P_1} \cdots \frac{P_T}{P_{T-1}}$ and apply ln to the product.

Exercise 4: Variance drag — Show that $\mathbb{E}[r] \approx \mathbb{E}[R] - \frac{1}{2}\sigma^2$

Using the Taylor expansion: $\ln(1+R) \approx R - R^2/2$, apply $\mathbb{E}[\cdot]$ to both sides.
Hint: $\mathbb{E}[R^2] = \mathrm{Var}(R) + (\mathbb{E}[R])^2 \approx \sigma^2$ for $\mathbb{E}[R] \approx 0$.

Exercise 5: Jensen's inequality — Derive $\mathbb{E}[\ln(1+R)] \leq \ln(1+\mathbb{E}[R])$

The function $f(x) = \ln(x)$ is concave. State Jensen's inequality and interpret: what does this imply for the relationship between the geometric and arithmetic return?

Exercises 3–5: Solutions

Solution Ex. 3 — Additivity

$$\frac{P_T}{P_0} = \prod_{t=1}^{T} \frac{P_t}{P_{t-1}} \quad \Rightarrow \quad \ln\left(\frac{P_T}{P_0}\right) = \sum_{t=1}^{T} \underbrace{\ln\left(\frac{P_t}{P_{t-1}}\right)}_{=r_t} \qquad \square$$

Solution Ex. 4 — Variance drag

$r = \ln(1+R) \approx R - \frac{R^2}{2}$. Apply $\mathbb{E}[\cdot]$: $\mathbb{E}[r] \approx \mathbb{E}[R] - \frac{1}{2}\mathbb{E}[R^2] = \mathbb{E}[R] - \frac{1}{2}\left(\sigma^2 + \mu^2\right) \approx \mathbb{E}[R] - \frac{1}{2}\sigma^2$ (since $\mu^2 \ll \sigma^2$) $\square$

Solution Ex. 5 — Jensen

- Jensen's inequality: if f is concave, then $\mathbb{E}[f(X)] \leq f(\mathbb{E}[X])$
- With $f = \ln$, $X = 1 + R$: $\mathbb{E}[\ln(1+R)] \leq \ln(1 + \mathbb{E}[R])$
- Interpretation: geometric return $<$ arithmetic return; the gap increases with σ $\square$

Why do we work with returns, not prices?

Prices: $P_t \sim I(1)$

- $\mathbb{E}[P_t]$, $\mathrm{Var}(P_t)$ are **not** constant
- Regression on prices $\Rightarrow$ **spurious regression** (Granger & Newbold, 1974)

Returns: $r_t \sim I(0)$

- Approximately stationary (mean ≈ 0, variance $\approx$ const.)
- Statistical methods are **valid**

Definition 2 (Weak stationarity (covariance stationarity))

A process $\{X_t\}$ is weakly stationary if, $\forall\, t, h$:

$$\mathbb{E}[X_t] = \mu, \qquad \mathrm{Var}(X_t) = \sigma^2 < \infty, \qquad \mathrm{Cov}(X_t, X_{t+h}) = \gamma(h).$$

Consequence

- $P_t \sim I(1)$: non-stationary (unit root)
- $r_t = \Delta \ln P_t \sim I(0)$: stationary
- Classical methods (OLS, t-tests, F-tests) **require stationarity**; on $I(1)$ series they produce spurious results

Random walk and non-stationarity

Random walk: $P_t = P_{t-1} + \varepsilon_t, \quad \varepsilon_t \overset{iid}{\sim} (0, \sigma^2)$

Why is P_t non-stationary?

- Iterating: $P_t = P_0 + \sum_{s=1}^{t} \varepsilon_s$
- $\mathrm{Var}(P_t) = t\,\sigma^2 \xrightarrow{t\to\infty} \infty$
- Variance **grows linearly** with t

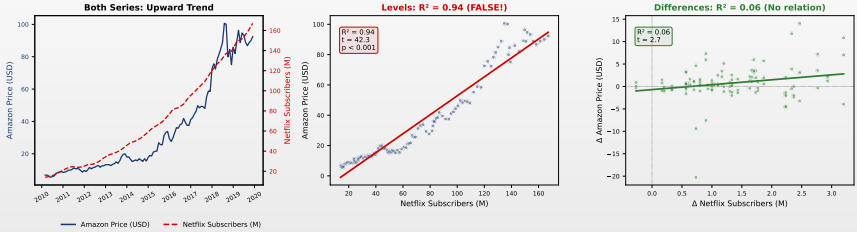
But $r_t = \Delta \ln P_t$ is stationary

- $\mathbb{E}[r_t] = \mu$ (constant)
- $\mathrm{Var}(r_t) = \sigma^2$ (constant)
- $\mathrm{Cov}(r_t, r_{t+h}) = \gamma(h)$ — depends only on h

Strict vs. weak

- **Strict**: the joint distribution $(X_{t_1}, \ldots, X_{t_k})$ is invariant to time shifts
- **Weak**: only the first two moments are invariant (sufficient for OLS, ACF)
- Strict $\Rightarrow$ weak (if $\mathbb{E}[X_t^2] < \infty$), but **not** conversely

Spurious regression: Amazon Price vs. Netflix Subscribers (2010–2020)



In levels (price vs. subscribers)

R^2	**0.94**
$\hat{\beta}$	0.63
t-stat	42.3
Conclusion	**Spurious!**

In differences (ΔP_t vs. ΔS_t)

R^2	**0.06**
$\hat{\beta}$	1.10
t-stat	2.7
Conclusion	**No relationship**

Lesson: A stock price and a subscriber count — **completely different** variables, but both trending $\Rightarrow R^2 = 0.94$ (Granger & Newbold, 1974). In differences: $R^2 = 0.06$.

SFM_ch1_stationarity

Python implementation: Download and compute

Download data + returns + descriptive statistics

```
import yfinance as yf
import numpy as np
import matplotlib.pyplot as plt
from statsmodels.tsa.stattools import adfuller
from scipy.stats import jarque_bera

ticker = "AAPL"
data = yf.download(ticker, start="2020-01-01", end="2024-01-01")
data['log_ret'] = np.log(data['Close'] / data['Close'].shift(1))
data['simple_ret'] = data['Close'].pct_change()
r = data['log_ret'].dropna()

print(f"Daily mean:     {r.mean():.6f}")
print(f"Annualised vol: {r.std() * np.sqrt(252):.4f}")
print(f"Skewness:       {r.skew():.4f}")
print(f"Kurtosis:       {r.kurtosis() + 3:.4f}")  # scipy: excess
```

SFM_ch1_returns

Python implementation: Statistical tests

ADF test (stationarity) and Jarque-Bera (normality)

```
# ADF test on log-prices vs. returns
adf_price = adfuller(np.log(data['Close'].dropna()))
adf_ret   = adfuller(r)
print(f"ADF log-price: stat={adf_price[0]:.3f}, p={adf_price[1]:.4f}")
print(f"ADF returns:   stat={adf_ret[0]:.3f},   p={adf_ret[1]:.4f}")

# Jarque-Bera test
jb_stat, jb_p = jarque_bera(r)
print(f"\nJarque-Bera: JB={jb_stat:.1f}, p-value={jb_p:.2e}")

# ACF on r_t and r_t^2
from statsmodels.tsa.stattools import acf
acf_r  = acf(r, nlags=30)
acf_r2 = acf(r**2, nlags=30)
```

- `adfuller` returns: (statistic, p-value, lags, nobs, critical values)
- $p < 0.05 \Rightarrow$ reject H_0 of unit root

SFM_ch1_stationarity

AI Exercise (I): Prompt Engineering

Task

- Write a prompt for an LLM (ChatGPT, Claude, etc.) that **fully** solves Exercise 1 from this seminar
- The prompt should include: input data, required formulas, desired output format

Example of a good prompt

- *"I have closing prices $P_0 = 100$, $P_1 = 110$, $P_2 = 99$, $P_3 = 120$. Compute simple returns R_t and log returns r_t for each period. Verify the additivity of log returns. Present the results in a table."*

Reflection questions

- Is the AI answer mathematically correct? Verify with manual calculation
- What happens if you change the wording? Do you get a different result?
- Did the AI explain **why** log returns are additive?

AI Exercise (II): Code Generation and Verification

Task

- Ask an LLM: *"Write Python code that downloads AAPL data for 2020–2024, computes log returns, tests stationarity with the ADF test, and displays a price vs. returns chart."*
- Run the generated code — does it work *as-is*?

Verification checklist

- Are the imports correct? (yfinance, numpy, statsmodels)
- Return formula: `np.log(P/P.shift(1))` or `pct_change()`?
  - Caution: `pct_change()` = **simple** returns, not log!
- ADF test: correct interpretation of the p-value?
- Does the plot have titles, axis labels, legend?

Pedagogical objective

- AI is a **tool**, not a substitute — generated code must **always be verified**
- Common error: confusion between simple and log returns

Discussion based on charts

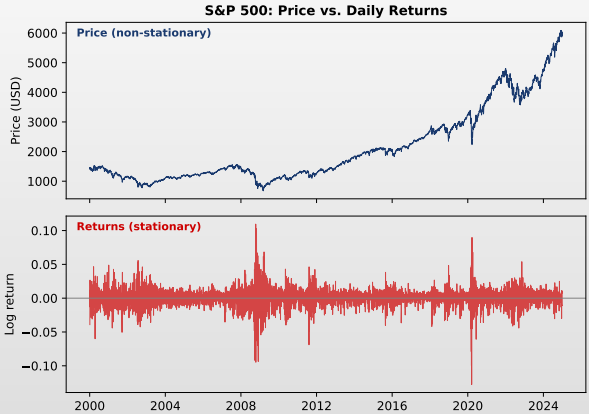
Questions for students — analysing the generated charts

1. Which series appears **stationary**? Which does not? Why?
2. Where do you observe **volatility clustering**?
   - Hint: look at March 2020 (COVID-19).
3. Are large returns followed by large returns (of any sign)?
4. Are there visibly "calm" periods alternating with "turbulent" ones?

Visually observable stylised facts

- **Clustering**: volatile periods come in groups
- **Fat tails**: a few extreme returns (far from the mean)
- **Mean** ≈ 0: returns fluctuate around zero
- These are the **stylised facts** of financial returns (Cont, 2001)

Case study: S&P 500 — Price vs. Returns



Data used

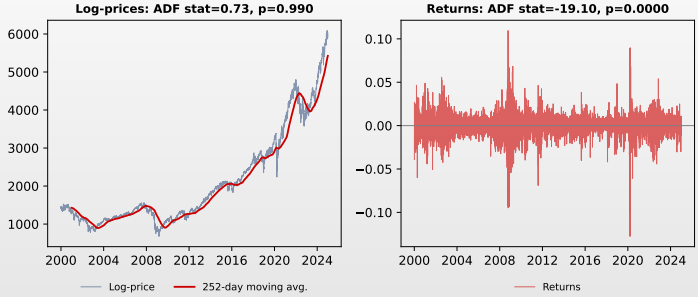
- S&P 500 Index
- Period: 2000–2024
- Frequency: daily

What do we observe?

- Price: upward trend, 2008 and 2020 crises visible
- Returns: oscillate around zero
- Volatility clustering evident during crisis periods

SFM_ch1_stationarity

Case study: S&P 500 — ADF Test



Log-prices

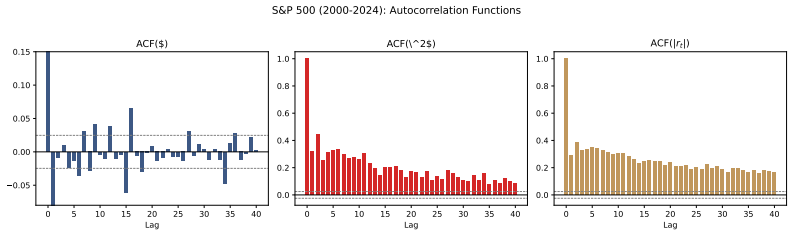
- H_0 is **not** rejected ($p = 0.98$)
- Non-stationary series, $I(1)$
- Unit root confirmed

Log-returns

- H_0 rejected at 1% ($p \approx 0$)
- Stationary series, $I(0)$
- Statistical methods are valid

SFM_ch1_stationarity

Case study: ACF — Returns vs. Squared Returns



$\mathrm{ACF}(r_t) \approx 0$

- Returns are **not** autocorrelated
- Consistent with the Efficient Market Hypothesis (EMH)
- Cannot predict r_{t+1} from r_t

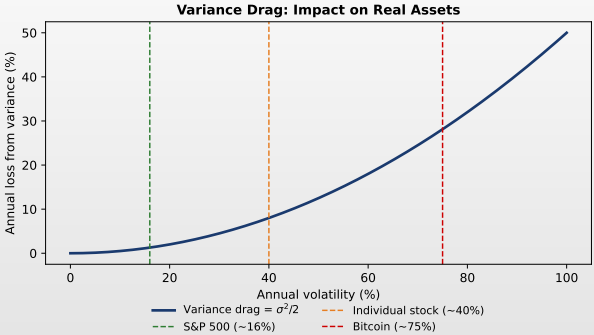
$\mathrm{ACF}(r_t^2) \neq 0$ — slow decay

- **Volatility clustering**: volatile periods persist
- Stylised fact #1 (Cont, 2001)
- Motivation for GARCH models: σ_t^2 depends on r_{t-1}^2

⚠ Returns are uncorrelated but **not** independent — dependence through higher-order moments

SFM_ch1_stylized_facts

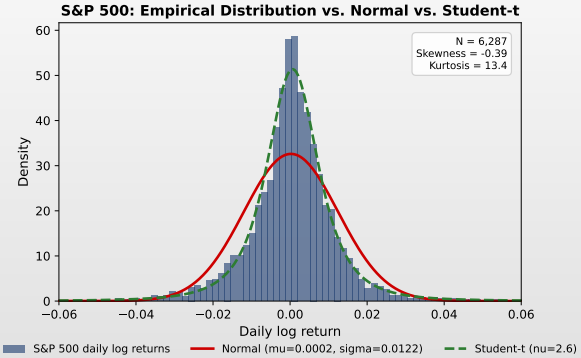
Case study: Variance Drag — Impact on real assets



- **S&P 500** ($\sigma \approx 16\%$): variance drag $\approx 1.3\%$/yr — negligible
- **Individual stocks** ($\sigma \approx 30$–40%): drag 4–8%/yr — significant
- **Bitcoin** ($\sigma \approx 75\%$): drag $\approx 28\%$/yr — devastating in the long run
- **Conclusion**: diversification reduces σ and hence reduces variance drag

SFM_ch1_variance_drag

Case study: S&P 500 — Empirical distribution vs. Normal



Real data: S&P 500 (2000–2024)

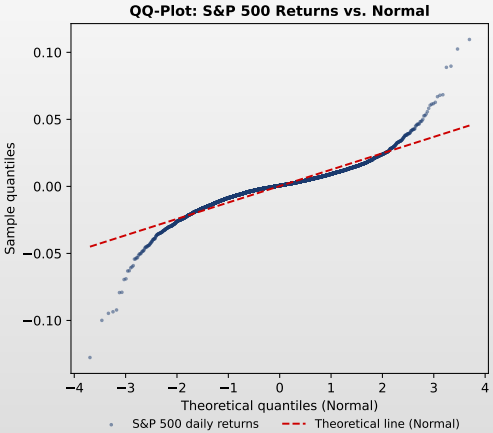
- $N = 6\,287$ daily log returns
- Empirical histogram vs. Normal curve vs. Student-t

What do we observe?

- Central peak **sharper** than Normal
- Tails **heavier** — crashes far more frequent
- $S = -0.39$, $K = 13.4$ (Normal: $S = 0$, $K = 3$)
- Student-t with $\nu \approx 3$ fits better

SFM_ch1_returns_analysis

Case study: QQ-Plot — Visual normality check



How to read a QQ-plot?

- Sample quantiles vs. Normal distribution quantiles
- If data are Normal $\Rightarrow$ points fall on the 45° line

What do we observe?

- **S-shape**: tails deviate from the line
  - Left tail: losses more extreme than Normal predicts
  - Right tail: gains more extreme
- The Jarque-Bera test formally confirms:
  - $JB = \frac{n}{6}\left(S^2 + \frac{\kappa^2}{4}\right) \sim \chi^2_2$

SFM_ch1_stylized_facts

Case study: Jarque-Bera Test — Formal rejection of normality

Definition 3 (Jarque-Bera statistic)

- H_0: $r_t \sim \mathcal{N}(\mu, \sigma^2)$
- $JB = \frac{n}{6}\left(S^2 + \frac{(K-3)^2}{4}\right) \xrightarrow{d} \chi^2_2$
- S = skewness, K = kurtosis, n = sample size

Computation on S&P 500 (2000–2024, $n = 6\,287$)

- $JB = \frac{6287}{6}\left((-0.39)^2 + (13.4-3)^2/4\right) = 1047.8 \times 27.19 = 28\,488$

Asset	n	S	K	**JB**	p-**value**
S&P 500	6 287	−0.39	13.4	28 488	$< 10^{-16}$
AAPL	2 516	−0.31	8.6	3 168	$< 10^{-16}$
Bitcoin	3 287	−0.15	7.4	2 675	$< 10^{-16}$

- Critical value: $\chi^2_{2,0.01} = 9.21$
- All $JB \gg 9.21 \Rightarrow$ reject H_0 at **any** significance level

Case study: Probability of extreme values

How likely are crashes under the normality assumption?

- Under $\mathcal{N}(\mu, \sigma^2)$, a 5σ event has prob. $\approx 2.87 \times 10^{-7}$
- **Black Monday** (19 Oct. 1987): S&P 500 -22.6% in one day ($> 20\sigma$)

Magnitude	**Normal prob.**	**Observed frequency**
Move $> 3\sigma$	0.27% (1 in 370 days)	~1 in 50–100 days
Move $> 4\sigma$	0.006% (1 in 16 000 days)	~1 in 500–1000 days
Move $> 5\sigma$	1 in 3.5 mil. days	~1 in 5000–10 000 days

Conclusion

- Returns are **leptokurtic**
- The Normal distribution dramatically underestimates tail risk

Practical implication

- VaR under normality **underestimates** extreme losses
- Empirically $\approx$ Student-t with $\nu \in [4, 8]$

Case study: Empirical moments — Cross-asset comparison

- Sample statistics for daily log returns, S&P 500 (2000–2024), others (2015–2024):

Asset	Mean	Std	Skewness	Kurtosis	Min	Max
S&P 500	+0.040%	1.00%	−0.39	13.4	−12.8%	+9.4%
AAPL	+0.058%	1.80%	−0.31	8.6	−13.0%	+11.9%
Bitcoin	+0.195%	3.90%	−0.15	7.4	−46.5%	+22.5%
EUR/USD	+0.000%	0.45%	+0.08	5.2	−2.7%	+3.1%

- All assets: excess kurtosis $\gg 0 \Rightarrow$ fat tails confirmed
- Stocks and indices: negative skewness $\Rightarrow$ crash risk dominates
- Bitcoin: highest volatility ($\sigma = 3.9\%$ daily $\approx 62\%$ annualised)
- EUR/USD: mean ≈ 0, nearly symmetric, but heavy tails

SFM_ch1_returns

Homework: Seminar 1

Analysis of a Romanian stock (BSE)

Choose a stock listed on the Bucharest Stock Exchange (e.g.: TLV, SNP, BRD, FP, SNG). Period: min. 3 years.

Computational requirements

1. Daily log returns (chart)
2. Descriptive statistics: $\hat{\mu}$, $\hat{\sigma}$, S, K
3. Annualised volatility ($\times\sqrt{252}$)
4. **ADF test** on log-prices and on returns (report statistic, p-value, conclusion)
5. **Jarque-Bera test**: compute JB, report p-value, interpret

Interpretation requirements

- ⊡ Are the returns stationary? (ADF)
- ⊡ Are they normally distributed? (JB)
- ⊡ Compare σ with S&P 500 ($\approx$ 16% annualised)
- ⊡ Compute variance drag — is it significant?

Deadline

- ⊡ Next seminar
- ⊡ Python code + charts + 1 page interpretation

What you should know after Seminar 1

Competence checklist

- ✓ **Simple vs. log difference**: simple are multiplicative, log are temporally additive
- ✓ **Return compounding**: $1 + R_{\text{total}} = \prod(1 + R_t)$ vs. $r_{\text{total}} = \sum r_t$
- ✓ **Variance drag**: $\mathbb{E}[r] \approx \mathbb{E}[R] - \sigma^2/2$; volatility destroys wealth
- ✓ **Stationarity rationale**: $P_t \sim I(1)$, $r_t \sim I(0)$; statistical methods require stationarity
- ✓ **Python implementation**: data download, compute r_t, price vs. returns charts

Cognitive level achieved

- ⊡ Operational calculation (manual + code)
- ⊡ Economic interpretation
- ⊡ Mathematics–finance connection
- ⊡ Beginning of critical thinking

Seminar structure — Chapter 1 (4 weeks)

Sem.	Topic	Content	Homework
1	Returns and stationarity (this seminar)	Compute R_t, r_t; variance drag; $I(1)$ vs $I(0)$; Python intro	Analyse BSE stock: returns
2	Volatility	CC, rolling vol, $\sqrt{T}$; EWMA; 20 vs 60 day comparison	30-day volatility for BSE stock
3	Stylised facts	Skew, kurtosis, QQ-plot; ACF($\lvert r_t \rvert$); normality	Jarque-Bera test; normality check
4	Performance and risk	Sharpe, MDD, Sortino; complete mini-project	Mini-project: full analysis

Quick quiz — Seminar 1 (15 minutes)

Answer on paper (no notes)

1. Write the **3 conditions** for weak stationarity (covariance stationarity).
2. Why is $\mathrm{ACF}(r_t) \approx 0$ but $\mathrm{ACF}(r_t^2) \neq 0$? What model does this motivate?
3. An asset has $\mathbb{E}[R] = 10\%$, $\sigma = 40\%$. Compute the variance drag and wealth after 20 years ($W_0 = 1$). Interpret.
4. An $R^2 = 0.92$ between the price of stock A and the price of stock B necessarily indicates a real relationship? Argue.
5. For a sample with $n = 1000$, $S = -0.5$, $K = 8$, compute the Jarque-Bera statistic and conclude.

Quick quiz — Answers

Answers

1. $\mathbb{E}[X_t] = \mu$; $\mathrm{Var}(X_t) = \sigma^2 < \infty$; $\mathrm{Cov}(X_t, X_{t+h}) = \gamma(h)$ — the first two moments are constant, autocovariance depends only on the lag h.
2. Returns are uncorrelated (EMH) but **not independent**: their magnitude is autocorrelated. $\mathrm{ACF}(r_t^2)$ captures **volatility clustering** $\Rightarrow$ GARCH motivation.
3. Drag $= \sigma^2/2 = 0.40^2/2 = 8\%/\mathrm{yr}$. $\mathbb{E}[r] = 10\% - 8\% = 2\%/\mathrm{yr}$. $W_{20} = e^{20 \times 0.02} = 1.49$. The 10% arithmetic return "melts" to just 2% geometric — extreme volatility destroys wealth.
4. **No**. Both prices can be $I(1)$ with common trends (spurious regression, Granger & Newbold 1974). Must test on returns (R^2 drops dramatically) or test for cointegration.
5. $JB = \frac{1000}{6}((-0.5)^2 + (8-3)^2/4) = 166.7 \times (0.25 + 6.25) = 1083.3$. $\chi^2_{2,0.01} = 9.21$. $1083.3 \gg 9.21 \Rightarrow$ reject H_0 of normality.

References and resources

Key references

- Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets*. 4th Ed., Springer.
- Cont, R. (2001). Empirical Properties of Asset Returns. *Quantitative Finance*, 1(2), 223–236.
- Dickey, D.A., Fuller, W.A. (1979). Distribution of the Estimators for AR Time Series with a Unit Root. *JASA*, 74, 427–431.
- Granger, C.W.J., Newbold, P. (1974). Spurious Regressions in Econometrics. *J. of Econometrics*, 2(2), 111–120.
- Mandelbrot, B. (1963). The Variation of Certain Speculative Prices. *J. of Business*, 36(4), 394–419.

Online resources

- **GitHub**: `https://github.com/QuantLet/Stat_fin_markets` — complete Python code
- **Quantlet**: `https://quantlet.com` **Quantinar**: `https://quantinar.com`

Thank you!

Questions?

Course materials available at: https://danpele.github.io/SFM/

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